

SHYMKENT, Kazakhstan

WMO: 383280

Lat: 42.3478N

Lon: 69.7100E

Elev: 604

StdP: 94.28

Time Zone: 6.00 (E06)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.0	-18.3	0.8	-14.7	-15.1	1.1	-10.5	8.4	5.1	7.2	3.4	1.6	120	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	38.2	18.8	36.9	18.5	35.2	18.1	20.6	33.2	19.8	32.5	19.2	32.1	3.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.9	25.1	15.7	12.0	24.4	14.8	11.3	23.9	62.5	33.4	59.6	32.2	57.2	32.1	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.8	5.9	-17.7	41.1	4.6	1.6	-21.0	42.2	-23.7	43.1	-26.2	44.0	-29.5	45.2		
(5)			-18.1	22.0	4.4	1.3	-21.2	22.9	-23.8	23.7	-26.3	24.4	-29.5	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	-0.1	2.0	9.0	14.5	20.1	24.8	26.9	26.3	20.6	13.5	6.8	1.0		
	DBStd	10.61	5.73	6.16	5.17	4.37	3.51	3.16	2.84	2.97	3.86	4.72	5.28	5.50		
	HDD10.0	1056	314	228	79	12	1	0	0	0	0	22	120	281		
	HDD18.3	2531	571	457	290	126	22	1	0	1	21	159	347	538		
	CDD10.0	2464	1	4	48	148	314	443	525	504	318	131	24	2		
	CDD18.3	898	0	0	1	12	77	195	267	246	88	10	1	0		
	CDH23.3	12099	0	0	26	168	925	2542	3731	3357	1172	167	13	0		
	CDH26.7	6392	0	0	3	36	348	1346	2171	1924	521	42	2	0		
(14) Wind	WSAvg	2.5	2.1	2.4	2.7	2.9	2.9	2.8	2.7	2.8	2.6	2.2	2.1	2.1		
(15) Precipitation	PrecAvg	600	75	84	81	72	56	24	12	4	9	40	72	71		
	PrecMax	826	155	194	145	157	142	68	81	23	56	95	163	155		
	PrecMin	394	23	12	37	7	2	0	0	0	0	1	8	16		
	PrecStd	111	34	39	28	32	34	16	16	5	12	31	47	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	19.8	26.2	30.0	34.2	38.8	41.1	40.0	36.0	30.8	25.2	17.0		
		MCWB	9.0	11.0	13.3	16.6	18.0	19.2	19.0	18.7	17.7	15.4	13.9	9.1		
	2%	DB	12.0	15.3	23.0	27.2	32.1	36.9	38.9	37.9	34.0	27.2	20.1	13.7		
		MCWB	7.0	9.0	12.7	16.1	17.5	18.6	19.2	18.6	17.0	14.1	11.5	7.9		
	5%	DB	9.5	12.7	20.1	25.2	30.3	35.1	37.0	36.2	32.0	25.0	17.0	10.9		
		MCWB	5.6	7.7	11.6	14.8	17.0	18.3	18.7	18.2	16.1	12.5	10.2	6.7		
	10%	DB	7.2	10.0	17.4	22.9	28.5	33.2	35.2	35.0	29.9	22.3	14.8	8.2		
		MCWB	4.4	6.2	10.6	14.0	16.8	17.8	18.3	17.9	15.2	12.0	9.0	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	11.6	14.8	18.8	20.5	21.5	22.1	21.5	19.2	16.1	14.5	10.1		
		MCDB	14.1	17.6	21.5	26.5	29.8	33.1	34.4	35.1	33.5	27.7	22.1	14.7		
	2%	WB	7.6	9.9	13.6	17.2	19.2	20.2	20.6	20.2	17.7	14.7	12.6	8.5		
		MCDB	11.3	14.4	20.9	24.5	28.7	32.1	33.9	33.7	31.8	25.0	18.3	12.9		
	5%	WB	6.1	8.2	12.4	16.0	18.2	19.4	19.8	19.4	16.6	13.7	11.0	6.9		
		MCDB	8.7	11.7	18.7	23.2	27.5	31.2	33.2	33.4	29.9	22.7	16.2	10.2		
	10%	WB	4.7	6.6	11.2	14.8	17.3	18.7	19.1	18.6	15.6	12.7	9.5	5.3		
		MCDB	7.0	9.7	16.8	21.2	26.4	30.5	32.8	32.8	28.4	20.7	13.6	8.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.2	11.6	12.1	13.7	15.1	15.4	15.9	15.7	13.6	10.7	9.6		
		MCDBR	12.0	13.2	15.7	15.9	16.5	17.2	17.4	17.6	18.1	17.2	14.2	12.7		
	5% WB	MCWBR	7.9	8.1	7.9	8.0	7.2	6.6	6.2	6.7	7.2	7.5	7.5	8.1		
		MCWBR	10.7	11.9	13.7	13.8	14.9	15.6	15.7	16.5	17.3	14.8	12.3	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.348	0.387	0.446	0.460	0.438	0.424	0.409	0.381	0.368	0.416	0.402	0.344		
	taud		2.094	2.025	1.958	1.987	2.109	2.170	2.217	2.278	2.267	2.042	2.033	2.106		
	Ebn at Noon		753	784	783	808	840	851	860	875	859	741	676	716		
	Edh at Noon		107	135	163	171	155	147	139	126	118	131	114	97		
(44) All-Sky Solar Radiation	RadAvg		1.78	2.66	3.76	5.18	6.59	7.46	7.43	6.69	5.23	3.31	1.97	1.44		
	RadStd		0.23	0.32	0.32	0.57	0.42	0.32	0.32	0.23	0.21	0.30	0.27	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	+1.40	N/A	N/A	N/A	N/A	+183	+133
(47)	Regional (9 neighbors)	N/A	N/A	N/A	+0.58	N/A	-0.58	N/A	N/A	+108	+78

Nomenclature: See separate page